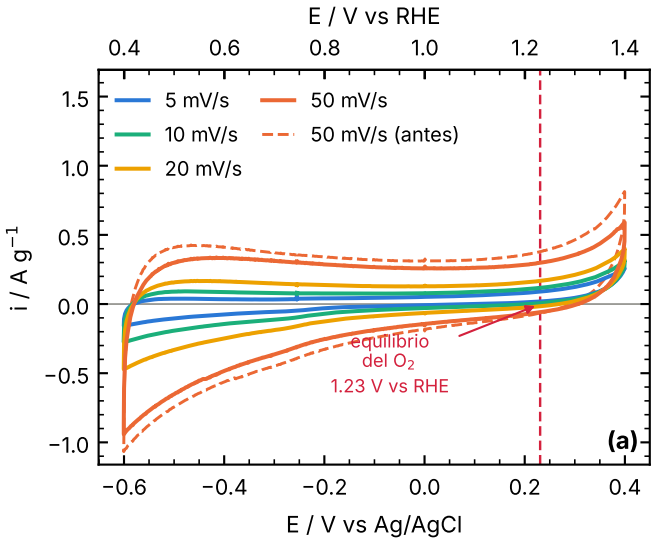
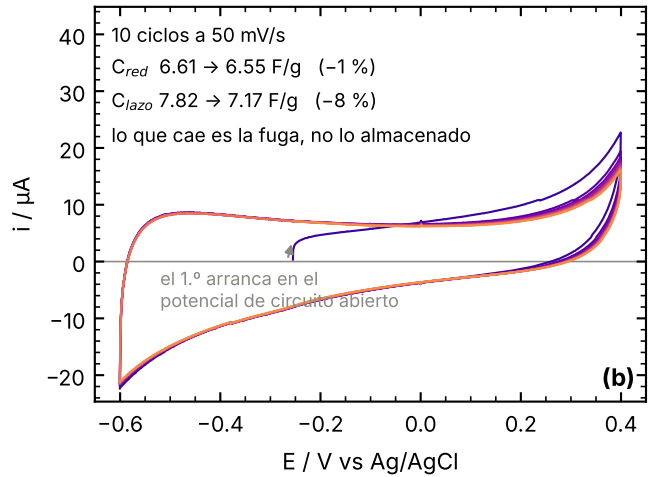
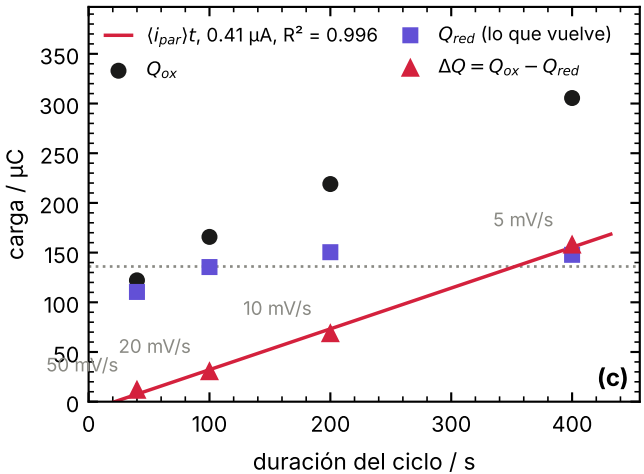
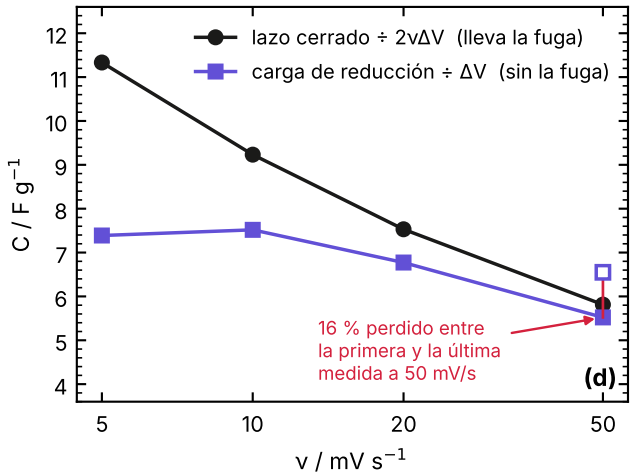

E / V vs RHE
0.4
0.6
0.8
1.0
1.2
1.4
5 mV/s
10 mV/s
20 mV/s
50 mV/s
50 mV/s (antes)
equilibrio
del O2
1.23 V vs RHE
(a)
i / A g−1
1.5
1.0
0.5
0.0
−0.5
−1.0
−0.6
−0.4
−0.2
0.0
0.2
0.4
E / V vs Ag/AgCl


10 ciclos a 50 mV/s
Cred 6.61 → 6.55 F/g (−1 %)
Clazo 7.82 → 7.17 F/g (−8 %)
lo que cae es la fuga, no lo almacenado
el 1.º arranca en el
potencial de circuito abierto
(b)
i / µA
40
30
20
10
0
−10
−20
−0.6
−0.4
−0.2
0.0
0.2
0.4
E / V vs Ag/AgCl


⟨ipar⟩t, 0.41 µA, R² = 0.996
Qox
Qred (lo que vuelve)
ΔQ = Qox − Qred
5 mV/s
10 mV/s
20 mV/s
50 mV/s
(c)
carga / µC
350
300
250
200
150
100
50
0
0
100
200
300
400
duración del ciclo / s


lazo cerrado ÷ 2vΔV (lleva la fuga)
carga de reducción ÷ ΔV (sin la fuga)
16 % perdido entre
la primera y la última
medida a 50 mV/s
(d)
C / F g−1
12
11
10
9
8
7
6
5
4
5
10
20
50
v / mV s−1